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Lagrange's polynomial

The Lagrange interpolation polynomial is a method used to find the polynomial of the lowest degree that passes through a given set of points. Given a set of $n + 1$ data points $(x_0, y_0), (x_1, y_1), \ldots, (x_n, y_n)$, the Lagrange interpolation polynomial $P(x)$ is constructed as a linear combination of Lagrange basis polynomials $L_i(x)$.
Steps to Construct the Polynomial:

1. Identify the Data Points: Collect the data points

$$(x_0, y_0), (x_1, y_1), ..., (x_n, y_n)$$

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2. Construct Lagrange Basis Polynomials: For each i from 0 to n, construct the basis polynomial $L_i(x)$:

$$L_i(x) = \prod_{\substack{0 \leq j \leq n \\ j \neq i}} \frac{x - x_j}{x_i - x_j}$$

3. Form the Interpolation Polynomial: Combine the basis polynomials weighted by the corresponding y_i values:

$$P(x) = \sum_{i=0}^{n} y_i L_i(x)$$

Problem
Consider the table

x	1	2	3
y	42	42.4	42.7

Using the Lagrange polynomial, find the value of $P(1.5)$.
The following steps are:

1. **Data Points**: $(x_0, y_0) = (1, 42)$, $(x_1, y_1) = (2, 42.4)$, $(x_2, y_2) = (3, 42.7)$
2. **Lagrange Polynomial**: The Lagrange interpolation polynomial $P(x)$ and the Lagrange basis polynomials $L_i(x)$ are given by:
$$P(x) = \sum_{i=0}^{n} y_i L_i(x) \qquad L_i(x) = \prod_{\substack{0 \le j \le n \\ j \ne i}} \frac{x - x_j}{x_i - x_j}$$
3. **Constructing the Basis Polynomials**: For $n = 2$, we have three basis polynomials:
$$L_0(x) = \frac{(x - x_1)(x - x_2)}{(x_0 - x_1)(x_0 - x_2)} = \frac{(x-2)(x-3)}{(1-2)(1-3)} = \frac{(x-2)(x-3)}{2}$$
$$L_1(x) = \frac{(x - x_0)(x - x_2)}{(x_1 - x_0)(x_1 - x_2)} = \frac{(x-1)(x-3)}{(2-1)(2-3)} = \frac{(x-1)(x-3)}{-1} = -(x-1)(x-3)$$
$$L_2(x) = \frac{(x - x_0)(x - x_1)}{(x_2 - x_0)(x_2 - x_1)} = \frac{(x-1)(x-2)}{(3-1)(3-2)} = \frac{(x-1)(x-2)}{2}$$
4. **Forming the Interpolation Polynomial**:
$$P(x) = y_0 L_0(x) + y_1 L_1(x) + y_2 L_2(x)$$
$$P(x) = 42\left(\frac{(x-2)(x-3)}{2}\right) - 42.4(x-1)(x-3) + 42.7\left(\frac{(x-1)(x-2)}{2}\right)$$
If this polynomial is simplified, the Lagrange interpolation polynomial $P(x)$ is:
$$P(x) = -0.05x^2 + 0.55x + 41.5$$

5. **Evaluate** $P(1.5)$: Substitute $x = 1.5$ into the polynomial:

$$L_0(1.5) = \frac{(1.5-2)(1.5-3)}{2} = \frac{(-0.5)(-1.5)}{2} = \frac{0.75}{2} = 0.375$$

$$L_1(1.5) = -(1.5-1)(1.5-3) = -(0.5)(-1.5) = -(-0.75) = 0.75$$

$$L_2(1.5) = \frac{(1.5-1)(1.5-2)}{2} = \frac{(0.5)(-0.5)}{2} = \frac{-0.25}{2} = -0.125$$

Therefore:

$$P(x) = y_0 L_0(x) + y_1 L_1(x) + y_2 L_2(x)$$

$$P(1.5) = 42 \cdot 0.375 + 42.4 \cdot 0.75 + 42.7 \cdot (-0.125)$$

$$P(1.5) = 15.75 + 31.8 - 5.3375 = 42.2125$$

So, the value of $P(1.5)$ using the Lagrange interpolation polynomial is 42.2125.